TELIPORT
Script Your Future
Season 4

TATA
TATA ELXSI

Case Study 2

Team Details

THINK. BUILD. CREATE. WIN

TELIPORT
Script Your Future
Season 4

Test a battery before you build it

VOLT-FORGE connects your battery design decisions to what they physically change

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PROBLEM UNDERSTANDING

Cells rarely share the load evenly

Say a Raspberry Pi 5 gateway must survive a **10-minute power cut at 25 W**, its full supply rating, at 23 °C. Four cells in parallel sounds easy, until the cell closest to the output takes more current at first. Ageing and packaging come later.

DESIRABILITY

You shouldn't need a lab to learn that

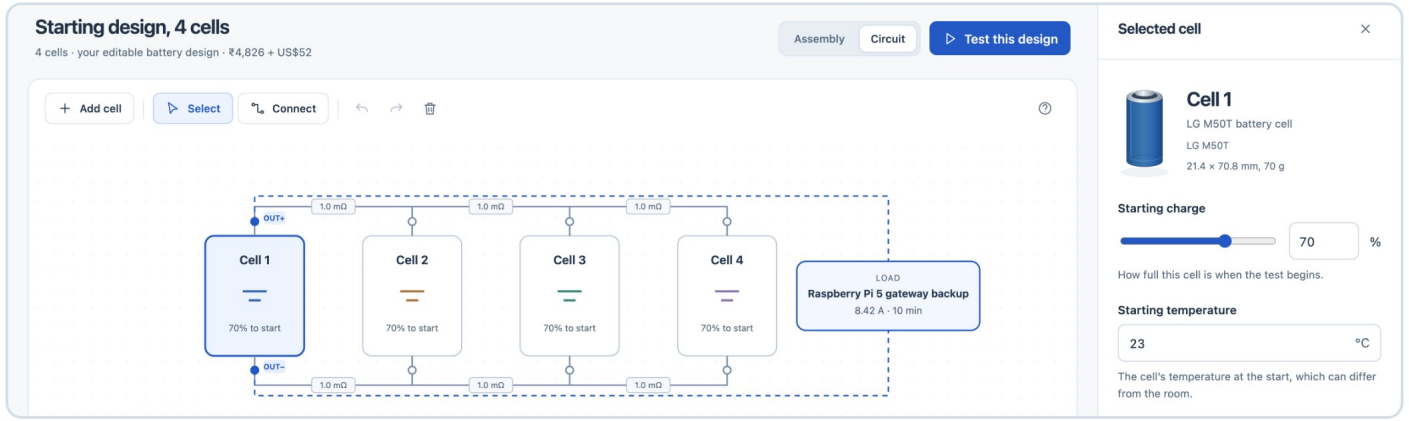
Today that takes a bench, a paid tool or your own code. Students can't afford that and engineers can't wait. Here you wire the circuit on screen and see which cell struggles **before spending a rupee**.

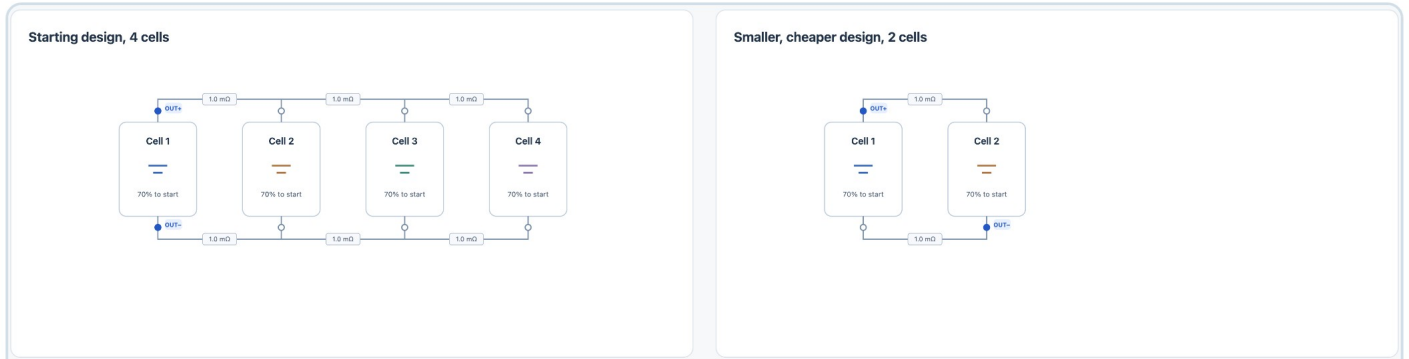
TECHNICAL FEASIBILITY

So how few cells can do this job?

We compared 3 and 2 cells against the 4-cell start. **Two cells** held the Pi for all 10 minutes in all 33 simulated tests, even with one weak cell. One cell was refused. It should fit a two-cell Pi UPS board, though we haven't built it yet, and the tool runs on a laptop.

PROTOTYPE & EVIDENCE The real Pi design, then 4 cells against 2 under the same load





Result	First test	Second test	Difference
Cells	4	2	−2
Listed cost, priced parts	₹4,826 + US$52	₹2,414 + US$46	
Cell mass	280 g	140 g	−140.00 g
Margin above voltage target	0.39 V	0.22 V	−0.16 V
Warmest cell	24.55 °C	27.37 °C	+2.82 °C

Result	First test	Second test	Difference
Largest cell current	2.64 A	4.21 A	+1.57 A
Energy delivered	5.26 Wh	5.08 Wh	−0.18 Wh
Lowest battery voltage	3.69 V	3.52 V	−0.16 V
Temperature difference	0.40 °C	0.00 °C	−0.40 °C
Dimensioned conductor mass	2.46 g	2.20 g	−0.26 g

4,006 simulated cases. 13 designs against weak cells, bad links and poor cooling. A full rerun matched exactly.

54 published four-cell experiments. All 194 dataset files verified intact, 2 tests compared with the model so far.

9 records kept closed. Saved for a final exam. 2 can test this cell, both at 40 °C.

ENGINEERING QUALITY

Why should you trust it?

Every run must balance its energy, which catches solver errors, not a wrong model. Against published data, voltage and temperature missed all 12 checks. Current sharing came closer but also missed: the nearest cell took **34%** early on, the model said **34%**. Over 10 minutes, real heating stayed within 1.13 times the model.

INNOVATION

You don't have to guess

Press **Find the smallest design**. It rebuilds the circuit at each size and tests it under normal conditions. For the Pi: 2 cells, not 4, at half the cell cost. Our study then stress-tested that.

BUSINESS VIABILITY



41.3 lakh TAM, engineering students

15.6 lakh SAM, electrical, electronics, mechanical

130 SOM, labs at ₹50,000, 5% assumed

₹35 crore

to build it all. Stage one is ₹11 crore: 130 labs and 31 teams at ₹5 lakh a year would repay it in 5 years, before running costs. Students use it free. No customer has seen these prices.

Simulated with an assumed 90% converter and 3.3 V floor. **Next:** a bench test with voltage, current and heat cutoffs, then student and engineer trials.